

It's On Sight – Project Documentation

Overview

This project is a 2D local multiplayer **Player vs Player (PvP)** arena game developed using Unity2D. Two players spawn into a battlefield and must fight until one of them is eliminated—either by running out of health or falling out of the stage boundaries.



Originally the idea from last semester was to have the player fight against some simple AI bots. But during this semester, I realized that this setup was kinda boring and limited in replayability and engagement. So, I changed it into a PvP Game, providing the player some competitive substance.

Gameplay

At the beginning of each match, two players would be spawned on two opposite sides of the map. The goal is simple: eliminate the other player. Players can do this by dealing damage through beating the brakes off each other or knocking the opponent off the edge of the stage.

- **Player 1 Controls:**
 - Movement: W, A, S, D
 - Attack 1: J
 - Attack 2: K
- **Player 2 Controls:**
 - Movement: Arrow keys
 - Attack 1: N

- Attack 2: M

Two character's traits:

- **Player 1** is a fast and agile fighter wearing white pants. Moves quickly and ideal for hit-



and-run strategies.

- **Player 2** is a powerful but slower boxer. Deals heavy damage and can withstand attacks



longer.

Environmental Mechanics

- **Turret:** A moving turret that shoots canon ball randomly
- **Jump Pads:** These spring players into the air, allowing for quick repositioning or surprise aerial attacks.
- **Slow-Motion Effects:** When a cannonball explodes near a player, the player will be pushed back with a slow-motion effect. Or when a player dies, the game enters slow motion to make the final blow look more dramatic.

Visual Design

The game's visual is quite simple. Since my main focus was the gameplay and the combat mechanics itself, so I just went for a minimalistic aesthetic. To enhance atmosphere, I just added a **raining effect** using Unity's particle system.

MoSCoW

Must Have

- Two functional and controllable characters with different traits
- Working health and damage system

- Win/lose condition (death or falling off map)
- Obstacle (turret, jump pads, falling platforms)

Should Have

- Slow-motion effects when cannonball hits the player and when the player's dead
- Character attack animations
- Attack effect (Knockback, stunt, impact vfx)
- UI showing player health or status (Health Bar)
- Camera shake when a player lands a blow

Could Have

- Background music and sound effects
- More detailed visual effects (dust trails when the player runs around)
- More characters with different abilities
- Multiple maps
- Player throwing objects at each others

Won't Have (for now)

Features outside the scope of the current version, which may be considered in future updates:

- Online multiplayer
- Character skins or customization
- Ranking system
- Combo attacks or skill chains

Post-Semester Goals

Even after this semester ends, I think I would like to continue developing the project whenever I have some free time. Here are the things I'd like to focus on when I do it:

- **Sound Design (SFX):**
To be honest, finding free sound and making some simple sound effect is quite easy so I

might add some impactful sound effects for attacks, explosions, platform breaking, and character grunts when taking damage in the future which would improve the game play experience quite a lot.

- **Visual Effects (VFX):**

I would consider myself not very good with vfx so I don't think I would be able to do much with this but when I do I would like to add some effects for dashing, jumping maybe adding trail or dust effect, and getting hit — also improving the slow-motion mechanic to enhance the highlight of the game.

- **UI Improvements:**

The current UI is quite simple yet usable. I'd like to design a better health bar system, maybe adding character selection, better pause menu, and maybe include some better round win/loss indicators, because as of now, I can see that the win loss in my current game doesn't seem that rewarding.

- **Adding Bots:**

Eventually, I'd like to reintroduce bots again, not as the main focus, but just as some fodder for the players to test out some moves or combos. Maybe adding some advanced bots which would be able to dodge turret fire and attack player to create a challenge.

- **Balancing & Replay Value:**

I'll continue tweaking the characters' stats, especially in terms of movement speed and damage, to ensure matches feel fair and competitive. I may also explore adding some power-ups like pickups or charging up through fighting.

Conclusion

This project represents my growth as both a developer and a designer. I moved from basic AI combat into PvP gameplay. While the game is still a work in progress, I'm pretty proud of the results and quite excited of its potential.

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